In [263…
```
import numpy as np
import pandas as pd
from scipy.stats import norm
```

Question 4

In [264…
```
df = pd.read_csv("data/ames_prices.csv")
```

In [265…
```
df["Central Air"] = (df["Central.Air"] == "Y").astype(int)
```

In [266…
```
df["Central Air"].value_counts()
```

Out[266…
```
Central Air
1    2734
0     196
Name: count, dtype: int64
```

In [267…
```
p_hat = sum(df["Central.Air"] == "Y") / len(df)
p_hat
```

Out[267…
```
0.9331058020477816
```

In [268…
```
len(df)
```

Out[268…
```
2930
```

Part A

Let's generate the boostrap distribution:

In [269…
```
rng = np.random.default_rng(seed=42)
```

In [270…
```
R = 2000
results = []

for _ in range(R):
    idx = rng.integers(0 , len(df), len(df))
    p_hat_star = df.loc[idx, "Central Air"].mean()
    results.append(p_hat_star)

results = np.array(results)
```

Part B

Now, let's estimate the standard error:

In [271…
```
standard_error = np.std(results, ddof=1)
standard_error
```

Out[271… np.float64(0.004590333440769273)

Part C

Now, let's construct the confidence intervals at 90%,95%, and99%$ using the centered-normal approach.

In [272…
```
# 90% CI
low = -1.645 * standard_error + p_hat
high = 1.645 * standard_error + p_hat

print(f"90% CI: [{low}, {high}]")
```

90% CI: [0.9255547035377162, 0.940656900557847]

In [273…
```
# 95% CI
low = -1.96 * standard_error + p_hat
high = 1.96 * standard_error + p_hat

print(f"95% CI: [{low}, {high}]")
```

95% CI: [0.9241087485038738, 0.9421028555916894]

In [274…
```
# 99% CI
low = -2.576 * standard_error + p_hat
high = 2.576 * standard_error + p_hat

print(f"99% CI: [{low}, {high}]")
```

99% CI: [0.92128110310436, 0.9449305009912032]

Part D

Now, let's perform the two-sided hypothesis test by computing the p-value. Note that each corresponding threshold of the hypothesis test is equivalent to one of the CIs above (a C% CI is equivalent to an $\alpha = 1 - C$ two-sided test), so we should expect them to line up accordingly.

In [275…
```
z = (p_hat - 0.9) / standard_error
p = 2 * norm.cdf(-abs(z))

print(z, p)
```

7.212069117626785 5.510798747816234e-13

Part E

0.000000000000551079874781623 4 is far smaller than any of our three signficance thresholds, so we would reject the null hypothesis at each of the signficance thresholds ($0.1, 0.05, 0.01$). This agrees with our corresponding confidence intervals, where all three intervals ($0.90, 0.95, 0.99$) do not contain the hypothesized value 0.9. A more thorough treatment is given in part f.

Part F

In context, our estimate, $\hat{p}$, is 0.933, meaning that approximately 93.3% of our sample Ames, Iowa houses contains central air. For confidence intervals, recall that a $C\%$ confidence interval may be interpreted as follows: if we were to repeat the same procedure (bootstrap to identify standard error and then construct a corresponding interval around our sample estimate using the associated critical value for our confidence) using many different random samples, $C\%$ of these intervals would contain the true population proportion. Since we only construct one of these intervals at three different confidences, we cannot be certain that the true population proportion falls within each of our intervals, though they still provide us with plausible values in which the true population proportion may take on. Thus, we have 90% confidence that the true proportion of houses with central air is between 0.925 and 0.941, 95% confidence that the true proportion of houses with central air is between 0.924 and 0.942, and 99% confidence that the true proportion of houses with central air is between 0.921 and 0.945. Since none of these intervals contain our hypothesized proportion of 0.90, the data suggests that our true population proportion is not equal to 0.90. This is confirmed by the corresponding hypothesis tests; as mentioned above, a C% CI is equivalent to an $\alpha = 1 - C$ two-sided hypothesis test, meaning that the p-value falls below our critical threshold exactly when the hypothesized value is not contained in the associated confidence interval. For instance, since 0.90 was not included in our 90% confidence interval, we should expect that our p-value will fall at or below the $\alpha = 0.1$ threshold; the similar statement applies for the pairs (95%, 0.05) and (99%, 0.01). As computed above, the standardized z-score is 7.21, and the p-value is $0.000000000000551079874781623 4$. The p-value represents the probability of obtaining a sample with a proportion atleast this extreme, assuming the null hypothesis is true. Since $0.000000000000551079874781623 4 \leq 0.1$, we reject the null hypothesis at $\alpha = 0.1$; a similar statement applies for $\alpha = 0.05$ and $\alpha = 0.01$, as anticipated. Thus, the data suggests (but does not prove) that the true proportion of Ames, Iowa houses with central air is not equal to 0.90. In particular, it seems that the true proportion may be higher, though it is poor form to conclude this given that we set-up a two-sided test; we are technically doubling our type 1 error because we did not declare this one-sided inequality before conducting the test.

Question 5

In [276…
```
corr_hat_matrix = np.corrcoef(np.arcsinh(df["area"]), np.arcsinh(df["price"]
corr_hat = corr_hat_matrix[0,1]
corr_hat
```

Out[276… np.float64(0.7231661530091358)

Part A

In [277…
```
R = 2000
results = []

for _ in range(R):
    idx = rng.integers(0 , len(df), len(df))
    corr_matrix = np.corrcoef(np.arcsinh(df.loc[idx, "area"]), np.arcsinh(df
    results.append(corr_matrix[0, 1])

results = np.array(results)
```

Part B

In [278…
```
standard_error = np.std(results, ddof=1)
standard_error
```

Out[278… np.float64(0.010353429300189908)

Part C

In [279…
```
# 90% CI
low = -1.645 * standard_error + corr_hat
high = 1.645 * standard_error + corr_hat

print(f"90% CI: [{low}, {high}]")
```

90% CI: [0.7061347618103234, 0.7401975442079481]

In [280…
```
# 95% CI
low = -1.96 * standard_error + corr_hat
high = 1.96 * standard_error + corr_hat

print(f"95% CI: [{low}, {high}]")
```

95% CI: [0.7028734315807635, 0.743458874437508]

In [281…
```
# 99% CI
low = -2.576 * standard_error + corr_hat
high = 2.576 * standard_error + corr_hat

print(f"99% CI: [{low}, {high}]")
```

99% CI: [0.6964957191318466, 0.749836586886425]

Part D

Now, let's perform the two-sided hypothesis test by computing the p-value. Note that each corresponding threshold of the hypothesis test is equivalent to one of the CIs above (a C% CI is equivalent to an $\alpha = 1 - C$ two-sided test), so we should expect them to line up accordingly.

In [282…
```
z = (corr_hat - 0) / standard_error
p = 2 * norm.cdf(-abs(z))

print(z, p)
```

```
69.84798292831063 0.0
```

A p-value of apporximately 0 is far smaller than any of our three signficance thresholds, so we would reject the null hypothesis at each of the signficance thresholds ($0.1, 0.05, 0.01$). This agrees with our corresponding confidence intervals, where all three intervals $(0.90, 0.95, 0.99)$ do not contain the hypothesized correlation value 0. A more thorough treatment is given in part f.

Part F

In context, our estimate, $\hat{\rho}$, is 0.723, meaning that the arcsinh tranformed varibles of area and height have a correlation of 0.723 within our sample of Ames Iowa homes; this is also our point estimate of the true correlation between arcsinh transformed area and arcsinh height in the overall population. For confidence intervals, recall that a $C\%$ confidence interval may be interpreted as follows: if we were to repeat the same procedure (bootstrap to identify standard error and then construct a corresponding interval around our sample estimate using the associated critical value for our confidence) using many different random samples, $C\%$ of these intervals would contain the true population correlation. Since we only construct one of these intervals at three different confidences, we cannot be certain that the true population correlation falls within each of our intervals, though they still provide us with plausible values in which the true population correlation may take on. Thus, we have 90% confidence that the true correlation of arcsinh(area) and arcsinh(price) for houses is between 0.706 and 0.740, 95% confidence that the true correlation of arcsinh(area) and arcsinh(price) for Ames, Iowa houses is between 0.703 and 0.743, and 99% confidence that the true correlation of arcsinh(area) and arcsinh(price) for houses is between 0.697 and 0.749. Since none of these intervals contain our hypothesized correlation of 0, the data suggests that our true population correlation for Ames, Iowa houses is not equal to 0. As computed above, the standardized z-score is 69.848, and the p-value is 0. The p-value represents the probability of obtaining a sample with a correlation atleast this extreme, assuming the null hypothesis is true. Since $0 \leq 0.1$, we reject the null hypothesis at $\alpha = 0.1$; a similar

statement applies for $\alpha = 0.05$ and $\alpha = 0.01$, as anticipated by the equivalence with the corresponding $1 - \alpha$ confidence interval. Thus, the data suggests (but does not prove) that the true correlation between arcsinh(area) and arcsinh(price) of Ames, Iowa houses is not equal to 0. In particular, it seems that the true correlation may be positive, though it is poor form to conclude this given that we set-up a two-sided test; we are technically doubling our type 1 error because we did not declare this one-sided inequality before conducting the test.

When we bootstrap, we are implicitly leveraging the fact that our sample (in this case, pairs of (arcsinh(area), arcsinh(height))) is an approximate stand-in for the population; this holds for a random sample from the true population. If we mix our pairs, we are destroying the integrity of our sample points, and we no longer have valid sample points from the population we are attempting to approximate. That is, instead of sampling from the emprical joint distribution (which is a stand-in for the true joint distribution whose correlation sampling distribution we are interested in), we are sampling from a joint empirical distribution that is the product of the marginal empirical distributions; these two distributions are equal if and only if the two variables are independent, which we do not know apriori. Thus, the distribution generated is mostly likely not approximating the intended sampling distribution.

Still, this regime can have application in other contexts. Moreover, resampling separately assumes that the two components of interest are independent of one another. As a result, the bootstrap distribution generated under this regime is a null, where the two variables are indpendent, and we can see where our sample correlation from the original data sits on it to suggest whether they are independent or not.

Question 7

In [283…
```
df = pd.read_csv("data/metabric.csv")
```

In [284…
```
mu_hat = df["Overall Survival (Months)"].mean()
```

In [285…
```
mu_hat
```

Out[285… `np.float64(129.09429138744105)`

Part A

In [286…
```
R = 2000
results = []

for _ in range(R):
    idx = rng.integers(0 , len(df), len(df))
    mu_hat_star = df.loc[idx, "Overall Survival (Months)"].mean()
```

```
    results.append(mu_hat_star)

results = np.array(results)
```

In [287…
```
results
```

Out[287…
```
array([129.94651278, 131.28240258, 128.44112683, ..., 129.98431373,
       133.34137503, 132.99339787], shape=(2000,))
```

Part B

In [288…
```
standard_error = np.std(results, ddof=1)
standard_error
```

Out[288…
```
np.float64(2.20091416851405)
```

Part C

In [289…
```
# 90% CI
low = -1.645 * standard_error + mu_hat
high = 1.645 * standard_error + mu_hat

print(f"90% CI: [{low}, {high}]")
```

```
90% CI: [125.47378758023544, 132.71479519464665]
```

In [290…
```
# 95% CI
low = -1.96 * standard_error + mu_hat
high = 1.96 * standard_error + mu_hat

print(f"95% CI: [{low}, {high}]")
```

```
95% CI: [124.78049961715351, 133.40808315772858]
```

In [291…
```
# 99% CI
low = -2.576 * standard_error + mu_hat
high = 2.576 * standard_error + mu_hat

print(f"99% CI: [{low}, {high}]")
```

```
99% CI: [123.42473648934886, 134.76384628553325]
```

Part D

Now, let's perform the two-sided hypothesis test by computing the p-value. Note that each corresponding threshold of the hypothesis test is equivalent to one of the CIs above (a C% CI is equivalent to an $\alpha = 1 - C$ two-sided test), so we should expect them to line up accordingly.

In [292…
```
z = (mu_hat - 120) / standard_error
p = 2 * norm.cdf(-abs(z))
```

```
print(z, p)
```

```
4.13205181626009 3.595393411803558e-05
```

Part E

3.595393411803558e-05 is far smaller than any of our three signficance thresholds, so we would reject the null hypothesis at each of the signficance thresholds ($0.1, 0.05, 0.01$). This agrees with our corresponding confidence intervals, where all three intervals ($0.90, 0.95, 0.99$) do not contain the hypothesized correlation value 0. A more thorough treatment is given in part f.

Part F

In context, our estimate, $\hat{\mu}$, is 129.094, meaning that the average recorded-time variable of our sample (formally $X = \min(T, C)$, where T is true death time and C is the time follow-up ends) is 129.094 months or 10.758 years. For confidence intervals, recall that a $C\%$ confidence interval may be interpreted as follows: if we were to repeat the same procedure (bootstrap to identify standard error and then construct a corresponding interval around our sample estimate using the associated critical value for our confidence) using many different random samples, $C\%$ of these intervals would contain the true population mean for our recorded-time variable. Since we only construct one of these intervals at three different confidences, we cannot be certain that the true population mean falls within each of our intervals, though they still provide us with plausible values in which the true population mean value may take on. Thus, we have 90% confidence that the true population mean for the recorded-time variable is between 125.474 (10.456) and 132.715 (11.060) months (years), 95% confidence that the true mean of the recorded time variable is between 124.780 (10.398) and 133.408 (11.117) months (years), and 99% confidence that the true mean of the recorded time variable is between 123.424 (10.285) and 134.764 (11.230) months (years). Since none of these intervals contain our hypothesized correlation of 120 months, the data suggests that our true population correlation is not equal to 120 months. As computed above, the standardized z-score is 4.13, and the p-value is 0.00003595393411803558. The p-value represents the probability of obtaining a sample with a mean atleast this extreme, assuming the null hypothesis is true. Since $0.00003595393411803558 \leq 0.1$, we reject the null hypothesis at $\alpha = 0.1$; a similar statement applies for $\alpha = 0.05$ and $\alpha = 0.01$, as anticipated by the equivalence with the corresponding $1 - \alpha$ confidence interval. Thus, the data suggests (but does not prove) that the true population mean for the recorded-time variable is not equal to 120 months or 10 years. In particular, it seems that the true mean may be higher, though it is poor form to conclude this given that we

set-up a two-sided test; we are technically doubling our type 1 error because we did not declare this one-sided inequality before conducting the test.

As made explicit throughout my explanation, the recorded-time variable is NOT death. Mathematically, it is the minimum of death time and the time follow-up ends. Since it is the minimum, it is pointwise less than death, and so using its expectation estimate as an estimate of death time systematically underestimates death time.

Question 8

In [293…
```
df = pd.read_csv("data/nhanes_data_17_18.csv")
```

```
/var/folders/wn/4vhmdq557_7f8325xdqxk2tr0000gn/T/ipykernel_3790/1514525784.py:1: DtypeWarning: Columns (142) have mixed types. Specify dtype option on import or set low_memory=False.
  df = pd.read_csv("data/nhanes_data_17_18.csv")
```

In [294…
```
df = df.loc[df["AgeInYearsAtScreening"] >= 18].reset_index()
```

In [295…
```
df
```

Out[295…

	index	SEQN	GeneralHealthCondition	EverBreastfedOrFedBreastmilk	AgeSto
0	2	93705.0	Good	NaN	
1	3	93706.0	Very good	NaN	
2	5	93708.0	Good	NaN	
3	6	93709.0	NaN	NaN	
4	7	93711.0	Very good	NaN	
...	...	...	...	...	
5528	8359	102949.0	Excellent	NaN	
5529	8361	102952.0	Very good	NaN	
5530	8362	102953.0	Fair or	NaN	
5531	8363	102954.0	Good	NaN	
5532	8365	102956.0	Good	NaN	

5533 rows × 199 columns

In [296…
```
mu_hat = df["SleepHoursWeekdaysOrWorkdays"].mean()
```

In [297…
```
mu_hat
```

Out[297…
```
np.float64(7.62124248496994)
```

Part A

In [298…
```
R = 2000
results = []

for _ in range(R):
    idx = rng.integers(0 , len(df), len(df))
    mu_hat_star = df.loc[idx, "SleepHoursWeekdaysOrWorkdays"].mean()
    results.append(mu_hat_star)

results = np.array(results)
```

Part B

Now, let's estimate the standard error

In [299…
```
standard_error = np.std(results, ddof=1)
standard_error
```

Out[299… np.float64(0.022697876273522776)

Part C

Now, let's construct the confidence intervals at 90%,95%, and99%$ using the centered-normal approach.

In [300…
```
# 90% CI
low = -1.645 * standard_error + mu_hat
high = 1.645 * standard_error + mu_hat

print(f"90% CI: [{low}, {high}]")
```

```
90% CI: [7.583904478499996, 7.658580491439885]
```

In [301…
```
# 95% CI
low = -1.96 * standard_error + mu_hat
high = 1.96 * standard_error + mu_hat

print(f"95% CI: [{low}, {high}]")
```

```
95% CI: [7.576754647473836, 7.665730322466045]
```

In [302…
```
# 99% CI
low = -2.576 * standard_error + mu_hat
high = 2.576 * standard_error + mu_hat

print(f"99% CI: [{low}, {high}]")
```

```
99% CI: [7.562772755689346, 7.679712214250535]
```

Part D

Now, let's perform the two-sided hypothesis test by computing the p-value. Note that each corresponding threshold of the hypothesis test is equivalent to one of the CIs above (a C% CI is equivalent to an $\alpha = 1 - C$ two-sided test), so we should expect them to line up accordingly.

In [303…
```
z = (mu_hat - 8) / standard_error
p = 2 * norm.cdf(-abs(z))

print(z, p)
```

```
-16.686914249853537 1.631902324600867e-62
```

Part E

1.631902324600867e-62 is far smaller than any of our three signficance thresholds, so we would reject the null hypothesis at each of the signficance thresholds ($0.1, 0.05, 0.01$). This agrees with our corresponding confidence intervals, where all three intervals ($0.90, 0.95, 0.99$) do not contain the hypothesized correlation value 0. A more thorough treatment is given in part f.

Part F

In context, our estimate, $\hat{\mu}$, is 7.621 hours, meaning the average weekday sleep-time for adults in our sample is about 7.621 hours. For confidence intervals, recall that a $C\%$ confidence interval may be interpreted as follows: if we were to repeat the same procedure (bootstrap to identify standard error and then construct a corresponding interval around our sample estimate using the associated critical value for our confidence) using many different random samples, $C\%$ of these intervals would contain the true population mean for our sleep variable. Since we only construct one of these intervals at three different confidences, we cannot be certain that the true population mean falls within each of our intervals, though they still provide us with plausible values in which the true population mean value may take on. Thus, we have 90% confidence that the true mean of sleep time for adults from the population sampled is between 7.584 and 7.659 hours, we have 95% confidence that the true mean of the sleep variable from the population sampled is between 7.577 and 7.666 hours, and 99% confidence that the true mean of the sleep variable from the population sampled is between 7.563 and 7.680 hours. As computed above, the standardized z-score is -16.687, and the p-value is $0.00000000000

extreme, assuming the null hypothesis is true. Since $0.00000000000